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Electronic device with below PCB thermal management

Abstract

Disclosed herein are electronic devices that utilized a thermal bus disposed on a backside of a printed circuit board (PCB) to route heat efficiently to a thermal management device disposed on the front side of the PCB, thus enhancing thermal regulation of integrated circuit (IC) devices mounted on the backside of the PCB.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10043730	12/2017	Refai-Ahmed	N/A	H01L 24/72
10147664	12/2017	Refai-Ahmed et al.	N/A	N/A
10262920	12/2018	Refai-Ahmed et al.	N/A	N/A
10529645	12/2019	Gandhi et al.	N/A	N/A
10720377	12/2019	Refai-Ahmed et al.	N/A	N/A
11145566	12/2020	Refai-Ahmed et al.	N/A	N/A
11195780	12/2020	Gandhi	N/A	H01L 25/0657
11246211	12/2021	Refai-Ahmed	N/A	H01L 23/4006
11328976	12/2021	Refai-Ahmed et al.	N/A	N/A
11330738	12/2021	Refai-Ahmed et al.	N/A	N/A
11355412	12/2021	Gandhi et al.	N/A	N/A
11373929	12/2021	Refai-Ahmed et al.	N/A	N/A
11488936	12/2021	Refai-Ahmed et al.	N/A	N/A
2020/0196482	12/2019	Chiang	N/A	H05K 7/20154
2021/0208647	12/2020	Macias	N/A	G06F 1/20
2021/0305127	12/2020	Refai-Ahmed et al.	N/A	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 18/113,585, filed Feb. 23, 2023 Entitled “Thermal Management System for Electronic Device”. cited by applicant

U.S. Appl. No. 18/102,065, filed Jan. 26, 2023 Entitled “Floating Heat Spreader With Guided Mechanism”. cited by applicant

U.S. Appl. No. 18/121,487, filed Mar. 14, 2023 Entitled “Frame for Coupling of a Thermal Management Device to a Printed Circuit Board”. cited by applicant

U.S. Appl. No. 17/688,803, filed Mar. 7, 2022 Entitled “Chip Package With Decoupled Thermal Management”. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) Embodiments of the present invention generally relate to electronic devices with one or more thermal buses disposed below one or more chip packages, and in particular, electronic devices having a thermal bus dispose below a printed circuit board that provides a heat transfer path from

below a chip package to a thermal management device disposed on top of the chip package that laterally by-passes the chip package.

BACKGROUND

(2) Electronic devices, such as tablets, computers, copiers, digital cameras, smart phones, control systems, automated teller machines, data centers, artificial intelligence system, and machine learning systems among others, often employ electronic and/or photonics components which leverage chip package assemblies for increased functionality and higher component density. Conventional chip packaging schemes often utilize a package substrate, often in conjunction with a through-silicon-via (TSV) interposer substrate and/or other such as FanOut and/or Silicon Bridging and/or substrate with glass and/or Si and/or organic core, to enable a plurality of integrated circuit (IC) dies to be mounted to a single package substrate. The IC dies are mounted to a die side (i.e., top surface) of the package substrate while a ball side (i.e., bottom surface) of the package substrate is mounted to a printed circuit board (PCB). The IC dies may include memory, logic or other IC devices. In many modern devices, one or more of the IC dies of the chip package are stacked one on top of the other to form a chip stack, which shortens routing and allows more IC dies to be utilized in the same foot print of the chip package.

(3) IC dies located deep on a chip stack are often difficult to cool. Heat from a bottom IC die is difficult to transfer through the IC die to a thermal management device disposed on top of the chip package, and the PCB generally has poor thermal conduction making heat flow out from under the stack not an effective option.

(4) Additionally, some IC devices that are integrated the supply of power or other functionality of the chip package are surface mounted below the PCB such that the routing between the IC dies and IC devices is reduced to improve efficiency and speed. These IC devices mounted below the PCB generally rely on ambient air movement or conduction through the PCB and chip package for cooling. However, since these are very inefficient heat transfer paths, the type, power and reliability of IC devices mounted below the PCB is limited.

(5) Therefore, a need exists for electronic devices having a heat management structures disposed below a chip package on the opposite side of the PCB.

SUMMARY

(6) Disclosed herein are electronic devices that utilized a thermal bus disposed on a backside of a printed circuit board (PCB) to route heat efficiently to a thermal management device disposed on the front side of the PCB, thus enhancing thermal regulation of integrated circuit (IC) devices mounted on the backside of the PCB. The thermal management device can be a thermal conductive material with or without exposed surfaces that extend into the surrounding fluid and/or solid state refrigerant. The thermal management device can also be a metal box that optionally has internal extended surface fill with a heat transfer fluid.

(7) In one example, an electronic device includes a PCB, a chip package, a thermal management device, an IC device and a thermal bus. The thermal management device sandwiches the chip package to a first side of the PCB. The IC device is mounted to a second side of the PCB that faces away from the heatsink. The thermal bus sandwiches the IC device to the second side of the PCB. A first conductive heat transfer path is defined between the thermal bus and the thermal management device. The first conductive heat transfer path is disposed laterally outward of the chip package.

(8) In some examples, the IC device is a heat generating device. The IC device may be an IC die, IC chiplet or a surface mounted circuit element. Examples of surface mounted circuit elements may include but are not limited to a voltage regulator, a voltage converter, a resistor, a capacitor, an inductor, and a transformer.

(9) In some examples, the thermal bus further includes one or both of an active or passive heat transfer device.

(10) In some examples, the thermal bus includes a phase change material disposed in a sealed

cavity.
(11) In some examples, the electronic device includes a thermal bus that is connected to the heatsink through a thermal via formed through the PCB. In other examples, the thermal bus is connected to thermal management device by a thermally conductive member that is not disposed through the PCB.
(12) In another example, an electronic device includes a PCB, a first chip package, a second chip package, a first thermal management device, a first heat generating IC device, a second heat generating IC device, and a first thermal bus. The first chip package and the second chip package are mounted to a first side of the PCB. The first thermal management device sandwiches the first chip package to the first side of the PCB. The first IC device is mounted to a second side of the PCB that faces away from the first thermal management device. The second heat generating IC device is mounted to the second side of the PCB. The first thermal bus sandwiches the first IC device to the second side of the PCB. The first thermal bus includes one or both of an active or passive heat transfer device. A first conductive heat transfer path is defined between the first thermal bus and the first thermal management device. The first conductive heat transfer path is disposed laterally outward of the first chip package.
(13) In some examples, a second thermal bus sandwiches the second IC device to the second side of the PCB. The second thermal bus includes one or both of an active or passive heat transfer device.
(14) In some examples, a lower junction plate is disposed between the first thermal bus and a second thermal bus. The lower junction plate is coupled to the first thermal bus and the second thermal bus by one or more passive heat transfer devices.
(15) In still other examples, a second thermal management device sandwiches the second chip package to the first side of the PCB. An upper junction plate is disposed between the first thermal management device and the second thermal management device. The upper junction plate is coupled to the first thermal management device and the second thermal management device by one or more passive heat transfer devices. A thermal via couples the upper junction plate to the lower junction plate through the PCB.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the above recited features of the present invention can be understood in detail, a more particular description of the invention, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments.
(2) FIG. **1** is a schematic sectional view of one example of an electronic device that includes at least one chip package mounted to a surface of a printed circuit board (PCB), where an opposite surface of the PCB includes an integrated circuit (IC) device interfaced with a thermal bus.
(3) FIG. **2** is another example of a chip package that may be utilized in the electronic device illustrated in FIG. **1**.
(4) FIG. **3** is a schematic sectional view of a portion of the electronic device illustrating one example of a portion of a conductive heat transfer path extending from the thermal bus through the PCB to a thermal management device.
(5) FIG. **4** is a schematic sectional view of a portion of the electronic device illustrating another example of a portion of a conductive heat transfer path extending from the thermal bus around the PCB to a thermal management device.

(6) FIG. **5** is a schematic partial sectional view of the electronic device of FIG. **1**, wherein the thermal bus is disposed on an opposite side of a backing plate.
(7) FIG. **6** is a schematic sectional view of another example of an electronic device that includes a plurality of chip packages mounted to a PCB and sharing a common a thermal bus.
(8) FIG. **7** is a schematic sectional view of another example of an electronic device that includes a plurality of chip packages mounted to a PCB and interfaced with different thermal buses.
(9) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements of one embodiment may be beneficially incorporated in other embodiments.

DETAILED DESCRIPTION

(10) As discussed above, IC dies located deep within a chip stack and IC devices that are located below the PCB are often difficult to cool. Disclosed herein is a thermal bus that is disposed below the chip package on the opposite side of the PCB. The thermal bus is configured to route heat from the side of the PCB opposite the chip package efficiently to a thermal management device residing above the chip package. Thus, the thermal bus improves heat conduction from the IC die closest the PCB. Additionally, when IC devices that present below the PCB, the thermal bus provides a good conductive heat transfer path to the thermal management device, which provides significantly heat removal from the IC device as compared to conventional ambient air convective cooling.
(11) In some examples, thermal conduction from the thermal bus is enhanced by the use of passive heat transfer mechanisms, such as heat pipes and vapor chambers disposed between the thermal bus and the thermal management device. In some examples, conductive vias are formed through the PCB to directly connect the thermal bus to the thermal management device.
(12) In some examples, IC devices such as power control circuit elements are conductively coupled to the thermal bus. The thermal bus provides excellent temperature control of these IC devices, enabling heat generating circuit elements to be safely mounted directly to the PCB below the chip package, when improves speed and processing performance without sacrificing reliability.
(13) Turning now to FIG. **1**, a schematic sectional view of an electronic device **180** that includes one or more chip packages **100** is illustrated. The chip package **100** includes at least one integrated circuit (IC) die **104** mounted to a top surface **166** of a package substrate **128** and a thermal bus **118** mounted to a bottom surface **168** of the package substrate **128**. The chip package **100** can be configured in heterogeneous integration with several chiplet and/or optical devices. Optionally, an interposer (not shown) may be disposed between the IC die **104** and the package substrate **128**. The package substrate **128** of the chip package **100** may be mounted on top surface **170** of a printed circuit board (PCB) **136** to form the electronic device **180**.
(14) A backing plate **114** is secured to a bottom surface **172** of the PCB **136** to provide additional rigidity and stiffness to the electronic device **180**. The backing plate **114** may be secured to the PCB **136** by clamps, fasteners, adhesives or other suitable technique. The backing plate **114** is typically fabricated from a metal or other rigid material. Optionally, the backing plate **114** and the thermal bus **118** may be integrated into a single structure or formed from the same mass of material.
(15) At least one integrated circuit (IC) device **102** is surface mounted to the bottom surface **172** of the PCB **136**. The backing plate **114** generally includes an aperture **116** through which the IC device **102** extends such that the IC device **102** does not interfere with the function of the backing plate **114**. Since the IC device **102** and the IC die **104** are disposed on opposite sides of the PCB **136**, additional space is available for other devices on the chip package side of the PCB **136**.
(16) The IC device **102** is coupled to functional circuitry **106** of the IC die **104**, rather than being formed within the IC die **104** or located on the package substrate **128** or other location within the chip package **100**. Thus, the IC device **102** is very close to the IC die **104** which enables excellent performance. Additionally, as the IC device **102** is not formed within the IC die **104**, space normally occupied by on-die circuitry is now free within the IC die **104** for additional IC devices,

improved power routing, and the like.

(17) The IC device **102** is generally a heat generating IC device. That is, the IC device **102** generates heat when in use. Examples of an IC device **102** include, but are not limited to, a voltage regulator, a voltage converter, a resistor, a capacitor, an inductor, and a transformer. In the example depicted in FIG. **1**, the IC device **102** is part of the power delivery circuitry that routes power from the PCB **136** to the functional circuitry **106** of the IC die **104**. Advantageously as the IC device **102** may be located very close to and even directly below the IC die **104**, routing between the functional circuitry **106** of the IC die **104** and the IC device **102** is relatively short, thus improving performance while reducing capacitive coupling. Although in FIG. **1** only a single IC device **102** is shown, as many IC devices **102** may be utilized as desired and as space permits.

(18) The IC device **102** is thermally regulated by use of at least one thermal bus **118**. The thermal bus **118** also enhances heat transfer from the IC dies **104** disposed near the package substrate **128**. One thermal bus **118** is shown in FIG. **1** although multiple thermal buses **118** may be utilized. The thermal bus **118** generally routes heat from the IC device **102** laterally below the bottom surface **172** of the PCB **136** and around (rather than through) the chip package **100** to a thermal management device **112** disposed on top of the IC die **104**, as shown by arrows that define a conductive heat transfer path **178**. The conductive heat transfer path **178** provides a more efficient heat transfer path between the IC device **102** and the thermal management device **112** as compared to conducting heat vertically through the PCB **136** and the package substrate **128** which are poor thermal conductors. Thus, the conductive heat transfer path **178** enables IC devices **102** to operate at higher power as compared to conventions designs that primarily rely convectional heat transfer to remove heat from the IC device **102**. The use of the below-PCB thermal bus **118** is particularly advantageously when the IC device **102** is configured as a component of the power delivery system of the IC die **104**, as the excellent temperature control provided by the thermal bus **118** allows high-powered IC devices **102** such as power transformers, power converters and the like, to be positioned directly below the IC dies **104** where fast response and diminished capacitive coupling result in robust performance of the electronic device **180**. Additionally, the thermal bus **118** improves heat conduction from the IC die(s) **102** closest the PCB **136**, thus improving performance and reliability of the chip package **100**. Different examples of how the conductive heat transfer path **178** may be configured are later described below.

(19) Continuing to describe the thermal bus **118**, the thermal bus **118** generally is fabricated from a thermally conductive material, with or without extended exposed surfaces that enhance heat transfer. The thermal bus **118** may include a cavity include a heat transfer fluid, refrigerant, or phase change material is disposed and/or circulated. Extended exposed surfaces, such as fins, baffles and the like, may project into the cavity to enhance heat transfer. The thermal bus **118** may also be a solid-state heat pump, i.e., thermoelectric device, or a metal box optionally having internal extended surface filled with fluid. The thermal bus **118** generally includes at least one or both of an active or passive heat transfer device. Examples of active heat transfer devices includes thermoelectric coolers, forced air heat exchangers, forced liquid heat exchangers, and the like. Examples of passive heat transfer devices includes vapor chambers, heat pipes, phase change materials, fins, and the like. Thermal interface material (TIM) **126** may be utilized between the IC device **102** and a first side **122** of the thermal bus **118** to enhance heat transfer therebetween. The TIM **126** may be a thermal grease, thermal gel, thermally conductive pad or other suitable material.

(20) In the example depicted in FIG. **1**, the thermal bus **118** is coupled to the thermal management device **112** of the electronic device **180** using a plurality of conductive members **176**. In one example, the conductive member **176** may be configured as a metal fastener, but in other examples described below, the conductive member **176** may be configured differently. The conductive member **176** includes a threaded end that engages a threaded hole **154** formed in the lower surface of the thermal management device **112** that faces the IC die **104**. The conductive member **176** extends through apertures **174**, **164**, **162** formed in the PCB **136**, backing plate **114** and thermal bus

118. The conductive member **176** includes a head **156** that retains a spring **158** between the head **156** and a bottom surface **124** of the thermal bus **118**. When the conductive member **176** is threaded into the threaded hole **154** formed in the thermal management device **112**, the spring **158** is compressed and urges the thermal bus **118** against both the backing plate **114** and, through the TIM **126**, the IC device **102**.

(21) Continuing to refer to FIG. **1**, the IC die **104** of the chip package **100** includes the functional circuitry **106**. The functional circuitry **106** may include block random access memory (BRAM), UltraRAM (URAM), digital signal processing (DSP) blocks, configurable logic elements (CLEs), and the like. The IC die **104** may be, but is not limited to, programmable logic devices, such as field programmable gate arrays (FPGA), memory devices, such as high band-width memory (HBM), optical devices, processors or other IC logic structures. The IC die **104** may optionally include optical devices such as photo-detectors, lasers, optical sources, and the like. In the example of FIG. **1**, the IC die **104** is a logic die having math processor (also known as math engine) circuitry for accelerating machine-learning math operations in hardware, such as self-driving cars, artificial intelligence and data-center neural-network applications.

(22) Optionally, the at least one IC die **104** may be a plurality of IC dies **104**. When a plurality of IC dies **104** are utilized, the IC dies **104** may be disposed in a vertical stack and/or disposed laterally side by side. It is contemplated that the IC dies **104** comprising the plurality of IC dies **104** may be the same or different types. Although only one IC die **104** is shown in FIG. **1**, the number of IC dies **104** disposed in the chip package **100** may vary from one to as many as can fit within the chip package **100**. Additionally, one or more of the IC dies **104** may optionally be configured as a chiplet.

(23) FIG. **2** example of a chip package **100** that may be utilized in the electronic device **180** illustrated in FIG. **1** that includes a chip stack **202**. The chip stack **202** includes a plurality of IC dies **104** that are stacked one on top of the other. Although four IC dies **104** are illustrated in the single chip stack **202** shown in FIG. **2**, the number of IC dies **104** comprising the chip stack **202** may vary from one to as many as can fit within the chip package **100**. Additionally, although only one chip stack **202** is shown in FIG. **2**, one or more additional IC dies **104** and/or one or more additional chip stacks **202** may be disposed adjacent the chip stack **202** shown in FIG. **2**.

(24) Referring back to FIG. **1**, the IC die **104** includes a die body **148** having a die bottom surface **152** and a die top surface **150** connected by the die sidewall **134**. Heat may be conducted from the die top surface **150** of the die body **148** to the thermal management device **112** through a layer of TIM **146**. The functional circuitry **106** is disposed within the die body **148** and includes routing that terminates on the die bottom surface **152** of the IC die **104**. In the example depicted in FIG. **1**, the functional circuitry **106** of the IC die **104** is electrically and mechanically coupled to package circuitry **182** formed in the package substrate **128** by interconnects **108**. In one example, the interconnects **108** are solder connections, such as solder bumps. The interconnects **108** may alternatively be formed by a hybrid bond layer or other suitable technique.

(25) The package substrate **128** generally includes at least an upper build-up layer disposed on a core. Optionally, a lower build-up layer may be disposed on the other side of the core from the upper build-up layer. The build-up layers includes a plurality of conductive layers and vias that are patterned to provide routing comprising a portion of the package circuitry **182**. One end of the package circuitry **182** formed in the upper build-up layer terminates at bond pads formed on a top surface **166** of the package substrate **128** where the package circuitry **182** connects to the contact pads of the IC die **104**. The other end of the package circuitry **182** formed in the package substrate **128** terminates at bond pads formed on the bottom surface **168** of the package substrate **128**.

(26) The package circuitry **182** is connected by solder balls **110** to circuitry **142** of the PCB **136** that terminates at the top surface **170** of the PCB **136**. The functional circuitry **106** of the IC die **104** is also coupled by the package circuitry **182** of the package substrate **128** and the circuitry **142** of the PCB **136** to the functional circuitry of the IC device **102**. The circuitry **142** of the PCB **136**

may also connect the functional circuitry of the IC device **102** to a power, ground or signal source exterior to the electronic device **180** through a PCB connector, such as a backplane connector, audio/video connector, card edge connector, and the like.
(27) FIG. **3** is a schematic sectional view of a portion of the electronic device **180** illustrating one example of a portion of a conductive heat transfer path **178** extending from the thermal bus **118** through the PCB **136** to the thermal management device **112**. In the example depicted in FIG. **3**, the thermal bus **118** is configured as a vapor chamber and includes a sealed cavity **330** partially filled with a phase change material **332**. Alternatively, the cavity **330** partially filled with the phase change material **332** shown in FIG. **3** may be part of one or more heat pipes. The phase change material **332** within the cavity **330** functions to transfer heat from the portion of the thermal bus **118** contacting the IC device **102** to the portion of the thermal bus **118** interfacing with the conductive member **176**, where the heat is then conducted through the conductive member **176** to thermal management device **112** (as shown in FIG. **1**).
(28) In the example depicted in FIG. **3**, the conductive member **176** is configured in three portions, such as a first portion **320**, a second portion **310**, and a thermal via **302**. The first portion **320**, the second portion **310**, and the thermal via **302** are generally fabricated from a material with high thermal conductivity, for example a metal such as copper, among others.
(29) The first portion **320** of the conductive member **176** includes a threaded section **322** and a head **324**. The first portion **320** extends through the apertures **164**, **162** formed in the backing plate **114** and the thermal bus **118**. The head **324** retains a spring **326** disposed between the head **324** and the bottom surface **124** of the thermal bus **118**.
(30) The thermal via **302** is disposed through the PCB **136** and is aligned with the apertures **162**, **164**. The thermal via **302** may be staked, plated or otherwise formed or disposed in the PCB **136**. The thermal via **302** includes a first threaded hole **306** and a second threaded hole **304**. The first threaded hole **306** is configured to receive the threaded section **322** defined at the end of the first portion **320** of the conductive member **176**. The threaded section **322** of the first portion **320** of the conductive member **176** is threaded into the first threaded hole **306** formed in the thermal via **302**, the spring **326** is compressed and urges the thermal bus **118** against both the backing plate **114** (and, through the TIM **126**, against the IC device **102** as shown in FIG. **1**).
(31) The second threaded hole **304** of the thermal via **302** is configured to receive a threaded section **312** defined at the end of the second portion **310** of the conductive member **176**. Similar to as shown in FIG. **1**, the opposite end of the second portion **310** includes threaded portion that engages a threaded hole formed in the thermal management device **112**.
(32) The conductive member **176** with thermal via **302** provides a very efficient conductive transfer path **178** through the PCB **136** directly to the thermal management device **112**. As discussed above, the efficient conductive transfer path **178** allows high powered IC devices **102** to be utilized on the bottom surface **172** of the PCB **136** that enhance the performance of the electronic device **180** while also enhancing cooling of IC dies **104** disposed close to the PCB **136**.
(33) FIG. **4** is a schematic sectional view of a portion of the electronic device **180** illustrating another example of a portion of the conductive heat transfer path **178** extending through a conductive member **410** from the thermal bus **118** around an outer edge **402** of the PCB **136** to the thermal management device **112**. The conductive member **410** may be utilized alternatively or in addition to any other conductive member, such as the conductive member **176**.
(34) The conductive member **410** is generally disposed proximate an outer edge **408** of the thermal bus **118**. The conductive member **410** has a first end **404** disposed in contact with the first side **122** of the thermal bus **118**. Optionally, TIM **126** (not shown in FIG. **4**) may be disposed between the first end **404** of the conductive member **410** and the thermal bus **118**. The first end **404** of the conductive member **410** may be coupled to the thermal bus **118** in any suitable manner, such as with fasteners, adhesives, clamping and the like. A second end **406** of the conductive member **410** is disposed in contact with the thermal management device **112**. Optionally, TIM **126** (not shown in

FIG. **4**) may be disposed between the second end **406** of the conductive member **410** and the thermal management device **112**. The second end **406** of the conductive member **410** may be coupled to the thermal management device **112** in any suitable manner, such as with fasteners, adhesives, clamping and the like. Alternatively, the conductive member **410** may be part of one of the thermal bus **118** or the thermal management device **112**. For example, the thermal management device **112** may include a leg (e.g., the conductive member **410**) that extends downward to the thermal bus **118**.

(35) In one example, the conductive member **410** is formed form a thermally conductive material, such as a metal. Suitable metals include copper, aluminum, and stainless steel, among others. The conductive member **410** may optionally include active and/or passive heat transfer devices. In the example depicted in FIG. **4**, the conductive member **410** includes a sealed cavity **412** that includes a phase change material **332**. The cavity **412** may form part of a heat pipe or vapor chamber. The phase change material **332** increases the efficiency of heat being transferred from the thermal bus **118** to the thermal management device **112**, with allows for high-powered IC devices **102** to be beneficially utilized below the PCB **136** as described above.

(36) FIG. **5** is a schematic partial sectional view of the electronic device **180** of FIG. **1**, wherein the thermal bus **118** is disposed on an opposite side of a backing plate **514**. In the example depicted in FIG. **5**, the thermal bus **118** include an aperture **502** through which a pad **504** projecting from the backing plate **514** extends into contact with the bottom surface **170** of the PCB **136**. The pad **504** generally has a height that is slightly greater that the combined height and thickness of the IC device **102** and the thermal bus **118** such that good contact is made between the IC device **102** and the thermal bus **118** through the TIM **126**.

(37) FIG. **6** is a schematic sectional view of another example of an electronic device **680** that includes a plurality of chip packages **100** mounted to a common PCB **136** and sharing a common a thermal bus **118**. The configuration of the electronic device **680** is substantially identical to that of the electronic device **180** described above except of the number of chip packages **100**. Although only two chip packages **100** are shown in the example depicted in FIG. **6**, as many chip packages **100** may be utilized as desired and as space permits. At least one IC device **102** is surface mounted on the bottom surface **170** of the PCB **136** below at least one or more of the chip packages **100**. In the example depicted in FIG. **6**, one or more IC devices **102** are surface mounted on the bottom surface **170** of the PCB **136** below at least one, or more, or all of the chip packages **100**.

(38) FIG. **7** is a schematic sectional view of another example of an electronic device **780** that includes a plurality of chip packages **100** mounted to a common PCB **136** and interfaced with different thermal buses **118**. At least one IC device **102** is respectively sandwiched between the PCB **136** and each of the thermal buses **118**. The electronic device **780** also includes a plurality of thermal management devices **112**. Alternatively, a single thermal management device **112** may be utilized within the electronic device **780** and coupled to each of the different thermal buses **118**. In the example depicted in FIG. **7**, each thermal bus **118** has a corresponding thermal management device **112**.

(39) A plurality of conductive members generally provide conductive heat transfer paths **178** between the thermal buses **118** and the thermal management devices **112**. The conductive members may be configured as any one or more of the conductive members **176**, **410** described above. The conductive heat transfer path **178** may also be routed between thermal buses **118** using heat pipes **724** or other passive or active heat transfer device. In the example, one end of the heat pipe **724** is disposed in contact with one of the thermal buses **118**, while the other end of the heat pipe **724** is disposed in conductive contact with another one of the thermal buses **118**. The other end of the heat pipe **724** may be disposed in direct contact with the other thermal bus **118**, or be disposed in direct contact with a lower junction plate **714**, as depicted in FIG. **7**. When the lower junction plate **714** is utilized, heat can transfer between the thermal buses **118** through the lower junction plate **714** via the heat pipes **724**. This allows heat from a hotter IC device **102** in contact with one thermal bus

118 to efficiently move heat to a thermal bus **118** that is in contact with a cooler IC device **102**, which more efficiently removes heat from all the IC devices **102**.
(40) In the example depicted in FIG. **7**, at least one or more conductive members **176** are disposed between at least two of the chip packages **100** of the electronic device **780**. The between-package conductive members **176** utilize a thermal via **302** formed through the PCB **136** such as described with reference to FIG. **3**. The between-package conductive member **176** may be directly coupled to one of the thermal management devices **112**, or coupled to an upper junction plate **712**. The upper junction plate **712** is coupled to the thermal management devices **112** via the heat pipes **724** or other active or passive heat transfer device. In the example depicted in FIG. **7**, one heat pipe **722** couples the upper junction plate **712** to one thermal management device **112**, while another heat pipe **722** couples the upper junction plate **712** to another one of the thermal management devices **112**. This allows heat removed from the IC devices **102** in contact with the thermal buses **118** to efficiently move heat via conduction to the thermal management device **112**, which improves the performance of the electronic device **780**.
(41) Thus, the thermal bus disclosed herein is configured to route heat from IC devices disposed on the side of the PCB opposite the chip package efficiently to a thermal management device residing above the chip package. The improved thermal management of the IC devices disposed on the bottom side of the PCB enables higher power IC devices to be utilized closer to the chip packages without significant performance degradation. The thermal bus provides an excellent conductive heat transfer path from the IC device to the thermal management device. In some examples, heat generating IC devices such as power control circuit elements are conductively coupled to the thermal bus directly below the chip packages which provide robust power delivery without capacitive coupling within the routings while also opening space on the chip package side of the PCB for additional processing components and circuit elements. Thus, speed and processing performance is enhanced without sacrificing reliability or compactness.
(42) While the foregoing is directed to embodiments of the present invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. An electronic device comprising: a printed circuit board (PCB); a chip package; a thermal management device sandwiching the chip package to a first side of the PCB; an integrated circuit (IC) device mounted to a second side of the PCB that faces away from the thermal management device; a backing plate disposed against the second side of the PCB and having an opening through which the IC device extends; a thermal bus sandwiching the IC device to the second side of the PCB, the thermal bus sandwiching the backing plate to the second side of the PCB; and a first conductive heat transfer path defined between the thermal bus and the thermal management device, the first conductive heat transfer path disposed laterally outward of the chip package.
2. The electronic device of claim 1, wherein the thermal bus further comprises: one or both of an active or passive heat transfer device.
3. The electronic device of claim 1, wherein the thermal bus further comprises: a phase change material disposed in a sealed cavity.
4. The electronic device of claim 1, wherein the thermal bus further comprises: a phase change material disposed in a heat pipe.
5. The electronic device of claim 1, wherein the thermal bus further comprises: a phase change material disposed in a vapor chamber.
6. The electronic device of claim 1, wherein the thermal bus is connected to the thermal management device through a thermal via formed through the PCB.
7. The electronic device of claim 1, wherein the thermal bus is connected to the thermal

management device by a thermally conductive member that is not disposed through the PCB.
8. The electronic device of claim 7, wherein the thermally conductive member has a phase change material disposed therein.
9. The electronic device of claim 1, wherein the thermal management device further comprises: one or both of an active or passive heat transfer device.
10. The electronic device of claim 1 further comprising: a thermal interface material disposed between the IC device and the thermal bus.
11. The electronic device of claim 1, wherein the IC device further comprises at least one of a voltage regulator, a voltage converter, a resistor, a capacitor, an inductor, or a transformer.
12. An electronic device comprising: a printed circuit board (PCB); a first chip package mounted to a first side of the PCB; a second chip package mounted to the first side of the PCB; a first thermal management device sandwiching the first chip package to the first side of the PCB; a first heat generating integrated circuit (IC) device mounted to a second side of the PCB that faces away from the first thermal management device; a second heat generating integrated circuit (IC) device mounted to the second side of the PCB; a backing plate disposed against the second side of the PCB and having a first opening through which the first heat generating IC device extends; a first thermal bus sandwiching the first IC device to the second side of the PCB, the first thermal bus sandwiching the backing plate to the second side of the PCB, the first thermal bus including one or both of an active or passive heat transfer device; and a first conductive heat transfer path defined between the first thermal bus and the first thermal management device, the first conductive heat transfer path disposed laterally outward of the first chip package.
13. The electronic device of claim 12, wherein the first thermal bus sandwiches the second IC device to the second side of the PCB, and wherein the first thermal management device sandwiches the second chip package to the first side of the PCB.
14. The electronic device of claim 12 further comprising: a second thermal bus sandwiching the second IC device to the second side of the PCB, the second thermal bus including one or both of an active or passive heat transfer device.
15. The electronic device of claim 14, wherein the first thermal management device sandwiches the second chip package to the first side of the PCB.
16. The electronic device of claim 14 further comprising: a second thermal bus sandwiching the second IC device to the second side of the PCB, the second thermal bus including one or both of an active or passive heat transfer device.
17. The electronic device of claim 16 further comprising: a lower junction plate disposed between the first thermal bus and the second thermal bus, the lower junction plate coupled to the first thermal bus and the second thermal bus by one or more passive heat transfer devices.
18. The electronic device of claim 17 further comprising: a second thermal management device sandwiching the second chip package to the first side of the PCB; an upper junction plate disposed between the first thermal management device and the second thermal management device, the upper junction plate coupled to the first thermal management device and the second thermal management device by one or more passive heat transfer devices; and a thermal via coupling the upper junction plate to the lower junction plate through the PCB.
19. The electronic device of claim 18, wherein the first thermal bus is connected to the first thermal management device by a thermally conductive member that is not disposed through the PCB.
20. The electronic device of claim 1, wherein the IC device is a heat generating device, and wherein the thermal bus further comprises one or both of an active or passive heat transfer device.